

When residue graphs fail to beat established stability predictors: a protein-held-out assessment and a structure-numbering audit

Computational protein redesign study, item 11a

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Abstract

We tested a small residue-graph neural network for single-amino-acid stability-change prediction on the experimental S669 benchmark. A corrected mapping derives the mutated sequence site from the only wild-type/mutant sequence difference and independently verifies the PDB residue number. All 669 rows yield structure graphs, covering 94 PDB accessions. A PDB-group holdout contains 66 variants across 24 unseen structures. The graph model has Spearman correlation 0.402 and mean absolute error (MAE) 1.422 kcal/mol; stored GeoDDG-Seq predictions have 0.403 and 1.310 on the same rows. The graph model did not outperform the existing predictor on MAE. A separate partial-export FireProtDB accession audit finds 1,470 direct residue-number-matched rows from 74 PDBs, but it is not a complete database census. Disease-target structures and published suppressor assays define limitations, not evidence that our model discovered therapeutic rescues. No predicted second-site mutation has been validated experimentally.

1 Question and scope

Disease-linked TP53, SOD1 and PTEN illustrate why structural context matters but do not supply evidence of successful compensatory redesign. TP53 core-domain structure 1TSR, SOD1 2C9V and PTEN 1D5R were fetched from the RCSB Protein Data Bank. We do not present predictions for these three targets as measured therapeutic rescues. The present technical question is narrower: how well does an accessible, reproducible residue-graph regressor generalize to unseen protein structures, and what are the dataset limitations for a future rescue screen? S669 is a benchmark of 669 annotated variants, with experimental DDG and third-party predictions. We did not rerun the third-party tools; their stored outputs are comparison columns. FireProtDB provides many experimental DDG records but its measurements and sign conventions cannot be merged with S669 by column name alone. The source ledger and exact files are in the repository. Our FireProtDB subset is partial because the unpaged bulk API request timed out; none of its subset rates are database-wide prevalence claims.

2 Data retrieval and provenance

S669 was obtained from the public GeoStab repository and its Zenodo release. The 94 PDB identifiers in S669 were fetched from RCSB and all 94 yielded at least one accepted graph. An earlier draft incorrectly treated the mutation-name residue label as a sequence offset; this excluded 249 rows that the corrected mapping now accepts. The corrected pipeline finds the sole sequence difference, aligns it to the structure and independently checks the label against the observed PDB residue number. It accepts 669 of 669 rows. The superseded 420-row analysis is isolated in clearly labeled legacy folders and is not cited as the current result. An incomplete FireProtDB CSV response (network timeout after approximately 300 MB) was filtered in chunks. This remains a partial snapshot. Of 1,112,013 observed export rows examined, 3,865 met numeric-DDG, PDB and substitution filters across 132 fetched structures. Direct wild-type amino-acid and numbered-residue checks matched 1,470 rows on 74 structures; these checks alone do not establish equivalent assay state or construct. The 94 S669 PDBs and 74

directly mapped FireProtDB PDBs overlap in 11 accessions, yielding 157 unique mutation-data structures; six separately studied disease-mutant structures raise the accession-backed used total to 163. The number of source studies remains two. Fetching a PDB without processing it is not counted.

3 Mutation and residue graph construction

For each mutation with an unambiguous single residue change, let the wild-type sequence be $s = s_1 \cdots s_L$ and mutant sequence s' . A site j is accepted only when

$$s_j \neq s'_j, \quad \sum_{i=1}^L \mathbf{1}[s_i \neq s'_i] = 1. \quad (1)$$

The sequence alignment maps position j to observed C-alpha residue position $\pi(j)$ in the PDB chain. A missing or mismatched C-alpha residue rejects the row. The mutation label is checked against the aligned PDB residue number rather than assumed to be the sequence offset; insertion codes and construct numbering require explicit scrutiny. We form a local vertex set

$$V_j = \text{nearest}_{\min(32, |V|)} \{i : \|r_i - r_{\pi(j)}\|_2 \leq 12 \text{ \AA}\}, \quad (2)$$

where r_i is the observed C-alpha coordinate and the mutation site is always included. Adjacency uses an 8-angstrom cutoff with self-loops:

$$A_{ik} = \mathbf{1}[\|r_i - r_k\|_2 \leq 8 \text{ \AA}], \quad A_{ii} = 1. \quad (3)$$

The symmetric degree-normalized adjacency is

$$\hat{A} = D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_k A_{ik}. \quad (4)$$

Each vertex has 42 features: 20 wild-type amino-acid indicators, 20 mutant indicators supported at the mutated vertex only, a site flag and site-relative radial distance divided by 12 angstrom. The feature choices cannot encode side-chain packing, hydration, ionization, or conformational ensembles. Edges are inferred from C-alpha proximity rather than physical interaction energy.

4 Model, loss and validation

We use an embedding $H^{(0)} = \text{ReLU}(XW_0 + b_0)$ and two residual graph layers

$$H^{(\ell+1)} = H^{(\ell)} + \text{ReLU}(\hat{A}H^{(\ell)}W_\ell + b_\ell), \quad \ell = 0, 1. \quad (5)$$

The readout concatenates a valid-vertex mean and mutated-site embedding:

$$z = \left[\frac{\sum_i m_i H_i}{\sum_i m_i}; \sum_i q_i H_i \right], \quad \widehat{\Delta\Delta G} = W_2 \text{ReLU}(W_1 z + b_1) + b_2. \quad (6)$$

Here m_i masks padding and q_i identifies the mutation site. A regression bug in an early prototype allowed padded nodes into the average; it was repaired and covered by an invariance test. Model training uses AdamW, 15 epochs, seed 27, and the Smooth L1 penalty

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{2}(\hat{y}_i - y_i)^2, & |\hat{y}_i - y_i| < 1, \\ |\hat{y}_i - y_i| - \frac{1}{2}, & \text{otherwise.} \end{cases} \quad (7)$$

The split is generated by a single fixed-seed GroupShuffleSplit with PDB accession as group. An independent exact-sequence audit finds 93 unique reference strings across 94 accepted PDBs, with 69 in training and 24 in testing and 0 exact train/test overlaps. Exact-string separation does not rule out homologs, fold-level leakage, or overlap with historical third-party predictor training sets. The hard non-overlap requirement is The hard non-overlap requirement is

$$\{p_i : i \in \mathcal{I}_{\text{train}}\} \cap \{p_i : i \in \mathcal{I}_{\text{test}}\} = \emptyset. \quad (8)$$

A standardized ridge regression on site residue, mutant, local amino-acid composition, radial feature and neighborhood size provides a simpler train-only baseline. The mean of the training labels is another baseline. Third-party published predictor columns from S669 are measured on the identical test set, where values exist. Their historical training overlap with these proteins is not established, so this is a common-row prediction comparison, *not* a leakage-controlled apples-to-apples training comparison.

5 Results and negative benchmark

All 669 S669 rows yielded graphs: 603 training variants from 70 PDBs and 66 test variants from 24 unseen PDBs. The two largest test structures account for 19.7% of test rows. The previous 420-row draft reflected a construct-numbering error and its quantitative outputs are not comparable with this corrected split. Reported metrics are Reported metrics are

$$\text{MAE} = N^{-1} \sum_i |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{N^{-1} \sum_i (y_i - \hat{y}_i)^2}. \quad (9)$$

Spearman correlation is the Pearson correlation of ranks,

$$\rho_s = \text{corr}(\text{rank}(y), \text{rank}(\hat{y})). \quad (10)$$

A constant train-mean predictor has undefined rank correlation; the JSON stores null, not zero. The MAE and Spearman values below were read from the committed `results/s669_group_holdout.json` after correcting the padding bug.

Method	Test variants	MAE	Spearman
Train mean	66	1.5450	undefined
Ridge	66	1.5035	0.3025
Residue GNN	66	1.4219	0.4024
GeoDDG-Seq, stored	66	1.3101	0.4030
GeoDDG-3D, stored	66	1.3213	0.4889
PremPS, stored	66	1.4831	0.2918
DDGun, stored	66	1.4827	0.3206
DDMut, stored	66	1.2897	0.4987

6 Mutation-label strata, not a new classifier

The fixed test set contains 46 negative and 20 nonnegative experimental DDG labels, describing the S669 column, not a clinical classification. Against stored GeoDDG-Seq predictions, GNN absolute error was lower on 31 of 66 variants and higher on 35. At PDB level, it had lower average error on 14 of 24 structures and higher error on 10. Four observed-label bins contain 26, 12, 18, 10 variants; GNN-minus-comparator MAE differences are +0.333, -0.164, +0.339, -0.543 kcal/mol. The positive-label tail has only 10 rows. These post hoc bins cannot serve as a prospective mutation-ranking gate. The aggregate GNN MAE remains worse, and no subgroup is claimed as a rescue discovery.

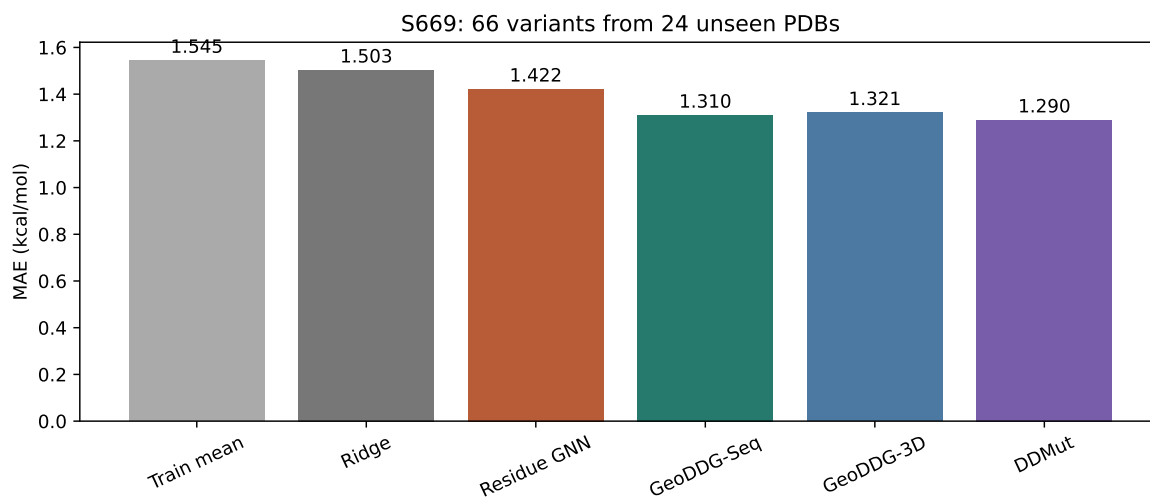


Figure 1: Error on the same S669 protein-group holdout. Smaller is better. Every displayed method has 66 stored predictions in this holdout. Third-party predictions are provided in S669 and not rerun here.

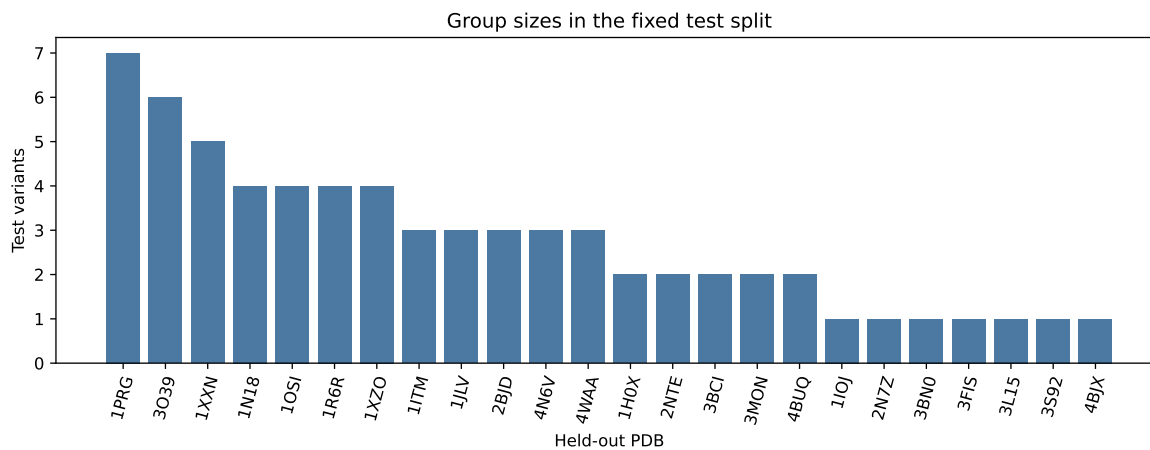


Figure 2: Sample concentration by test-set PDB accession. Reporting only row-weighted performance hides the concentration in two proteins.

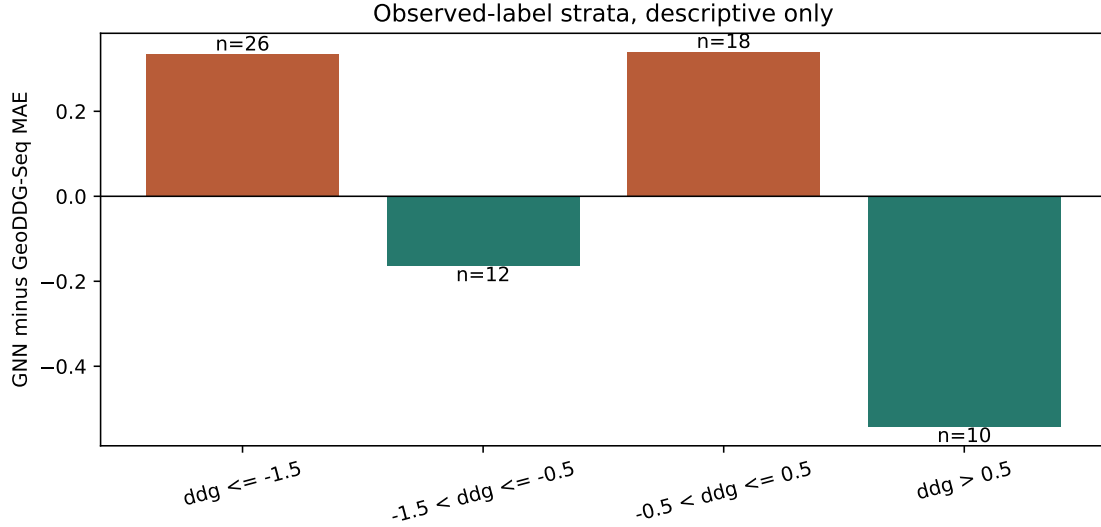


Figure 3: Observed-label error strata; labels are known only after measurement and group samples are small.

7 Group-weighting sensitivity

Define the per-mutation absolute-error difference against a published comparator as

$$\delta_i = |y_i - \hat{y}_i^{\text{GNN}}| - |y_i - \hat{y}_i^{\text{GeoDDG-Seq}}|. \quad (11)$$

Positive values favor the comparator. A row-weighted average and an equal-protein average differ:

$$\bar{\delta}_{\text{row}} = \frac{1}{N} \sum_i \delta_i, \quad \bar{\delta}_{\text{protein}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{n_k} \sum_{i:p_i=k} \delta_i. \quad (12)$$

The corrected audit finds $\bar{\delta}_{\text{row}} = +0.1118$ and $\bar{\delta}_{\text{protein}} = -0.0413$ kcal/mol across $K = 24$ groups. The signs differ under unequal weighting, which makes a single superiority claim unsafe. An exploratory group-bootstrap resamples PDBs with replacement and records the equally weighted group average:

$$\delta^{*(b)} = K^{-1} \sum_{j=1}^K \bar{\delta}_{g_j^{(b)}}, \quad g_j^{(b)} \sim \text{Unif}\{1, \dots, K\}. \quad (13)$$

The 2.5% and 97.5% protein-equal bootstrap quantiles are $[-0.4337, +0.3516]$ kcal/mol; they include zero. A naive row-bootstrap interval is $[-0.1273, +0.3436]$, also crossing zero. These descriptive intervals condition on one fixed partition; resampling related variants within a protein is not a new validation cohort. The reversal between row and protein weighting is a warning about small and uneven groups, not evidence for new-protein superiority.

8 Leave-one-structure-out sensitivity

The fixed benchmark split contains 66 test variants across 24 PDB groups. For a separate ridge-baseline stress test, each of the 94 graph-usable PDB structures is held out once; scaling and ridge fitting use only the remaining structures. The GNN is not retrained in these folds. Across 669 out-of-fold rows, ridge MAE is 1.1812 kcal/mol row-weighted and 1.4151 protein-equal. Train-mean MAE is 1.2492 and 1.4632, respectively. Ridge beats train

mean in 48 of 94 held-out structures. Median group size is 3, and 27 groups contain one variant. These are ridge sensitivity results, not GNN cross-validation, and exact-sequence checks do not exclude all homologs.

$$\text{MAE}_{\text{protein equal}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{n_k} \sum_{i:p_i=k} |y_i - \hat{y}_i^{(-k)}|. \quad (14)$$

Here the superscript $(-k)$ denotes a predictor fitted without PDB k . No protein-family homology clustering was attempted, so correlated structures may still straddle train and test. The ridge alpha of 100 is fixed in advance rather than tuned on the held-out proteins. A full model comparison would repeat GNN training across these 94 folds, multiple seeds, and independent structure families; no such claim is made here.

9 Sparse-PDB sensitivity of the negative comparison

Seven of the 24 held-out PDBs have a single test variant. Removing those groups leaves 17 PDBs and 59 variants; 9 PDBs then have lower GNN error. Requiring at least five variants retains just 3 PDBs and 18 rows, with no lower-error GNN group. The row-weighted GNN-minus-comparator MAE differences at thresholds 1, 2 and 5 are +0.112, +0.192 and +0.288 kcal/mol, respectively. This changes which proteins are represented after seeing the fixed test set; it is descriptive sensitivity, not model selection or an independent benchmark.

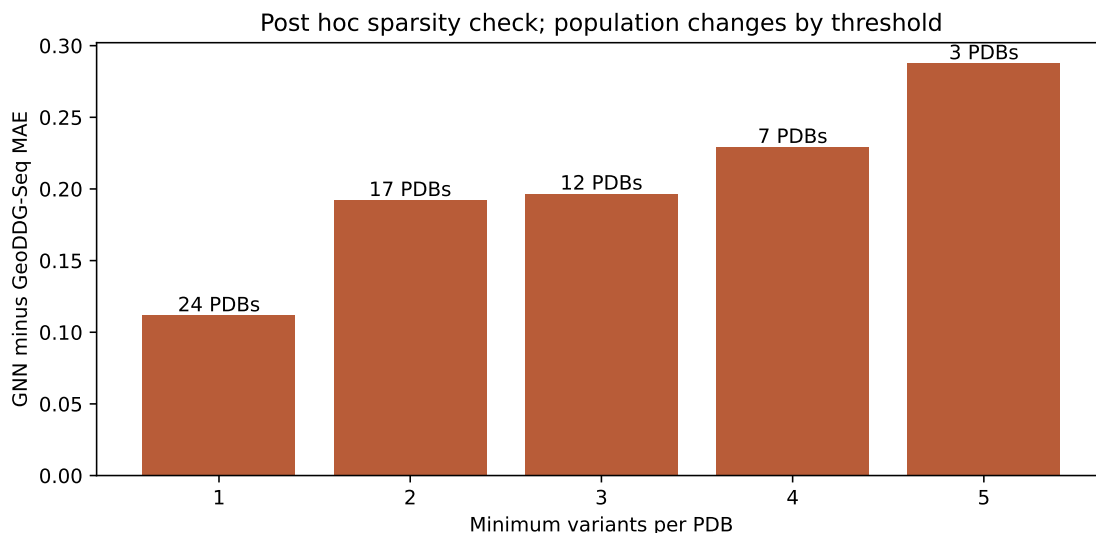


Figure 4: Row-weighted comparator gap after group-size thresholds; the sample changes each time.

10 FireProtDB as a separate audit, not pooled training

For each row in the observed FireProtDB subset with four-character PDB accession and single substitution, we compare the annotated wild-type residue and its original integer position with every resolved chain of its indicated structure. This directly matches 1,470 rows on 74 structures, from 3,865 retained experimental rows across 132 downloaded structures. A direct match is a screen, not an alignment guarantee. When one PDB and substitution have repeated DDG readings of both signs, we call the pair *sign-discordant* rather than contradictory. We count 147 PDB/substitution pairs of this kind; the committed subset audit finds 88 pairs with at least one sign disagreement even under the same recorded pH and experimental temperature. This is not proof that the same physical experiment was repeated: the database may combine distinct constructs, publication measures or metadata omissions. The FireProtDB official help page explicitly classifies negative DDG as stabilizing and

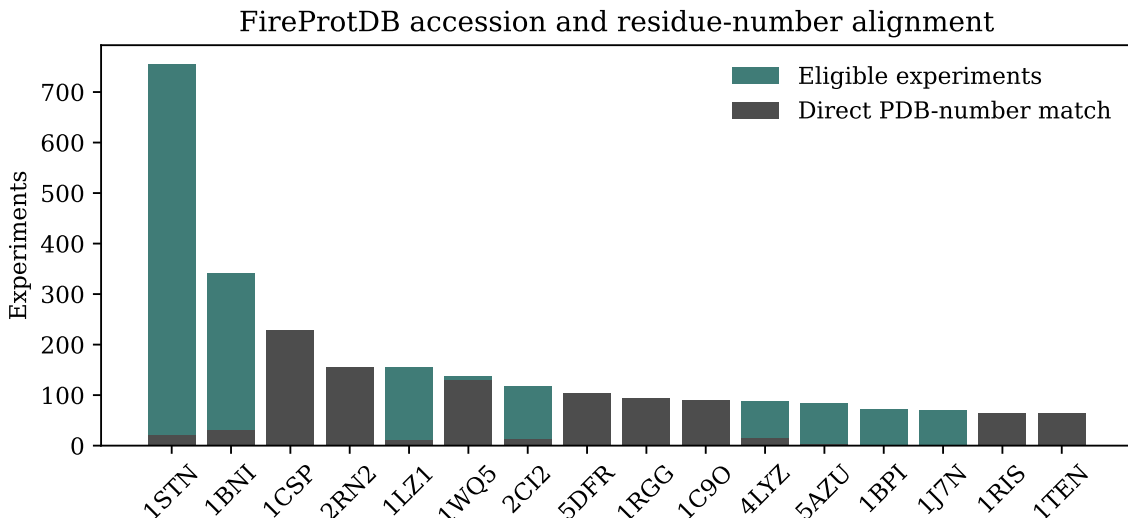


Figure 5: Among the largest accession groups, eligible experimental rows and direct PDB-residue-number matches differ sharply. Gray indicates direct matches, not independent experimental validation.

positive DDG as destabilizing. That definition is grounded in the site documentation; the S669 sign and physical-state definition still need independent source confirmation. We do not equate conventions from column names alone. If a hypothetical source uses $\Delta\Delta G_1 = G_{\text{mut}} - G_{\text{wt}}$ and another reports folding stability $\Delta\Delta G_2 = \Delta G_{\text{fold,mut}} - \Delta G_{\text{fold,wt}}$ with reversed folding free-energy convention, then

$$\Delta\Delta G_1 = -\Delta\Delta G_2 \quad (15)$$

may hold only when the underlying state definitions match. We have not demonstrated that equality for these datasets. Training on pooled signed labels would risk manufacturing a false rescue ranking, so the FireProtDB material remains an audit, not training labels in the S669 model.

11 Diagnostic figures and uncertainty interpretation

Figure 6 plots the mean absolute-error difference by PDB for the single fixed test partition. Positive values mean the GNN had larger errors than the stored GeoDDG-Seq predictions for that protein. Individual bars are not independent proof of a special mechanism; some proteins have one labeled mutation. Figure 7 compares the number of variants in each PDB with the ridge MAE obtained when that PDB alone was left out. Its logarithmic horizontal axis shows uneven group sizes; no one figure establishes a general population error estimate. We include this diagnostic so no reader mistakes 669 mutation rows for 669 independent protein generalization trials. Among the 753 repeated PDB/substitution groups in the observed partial FireProtDB snapshot, 675 have a nonzero DDG span. The median span over all 753 groups is 0.50 kcal/mol, the 90th percentile 3.68, and the 95th percentile 5.10. Figure 8 retains the full measured spread rather than pretending that one chosen number represents a unique mutation effect. It is a record-level variation statistic, not repeat-assay precision: two records sharing PDB and mutation may differ in unrecorded construct, source paper, folding state, buffer, or assay design. We therefore do not use a high-span group as a clinical variant training target without resolving provenance first.

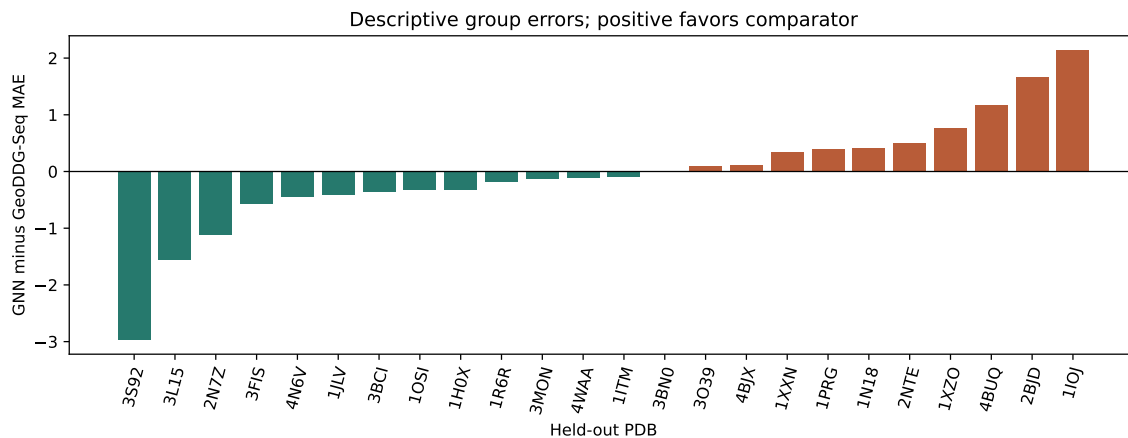


Figure 6: Protein-wise difference in mean absolute error for the GNN and an existing predictor on one fixed holdout. Red indicates poorer GNN error; teal indicates lower GNN error for that accession. Small groups have high uncertainty.

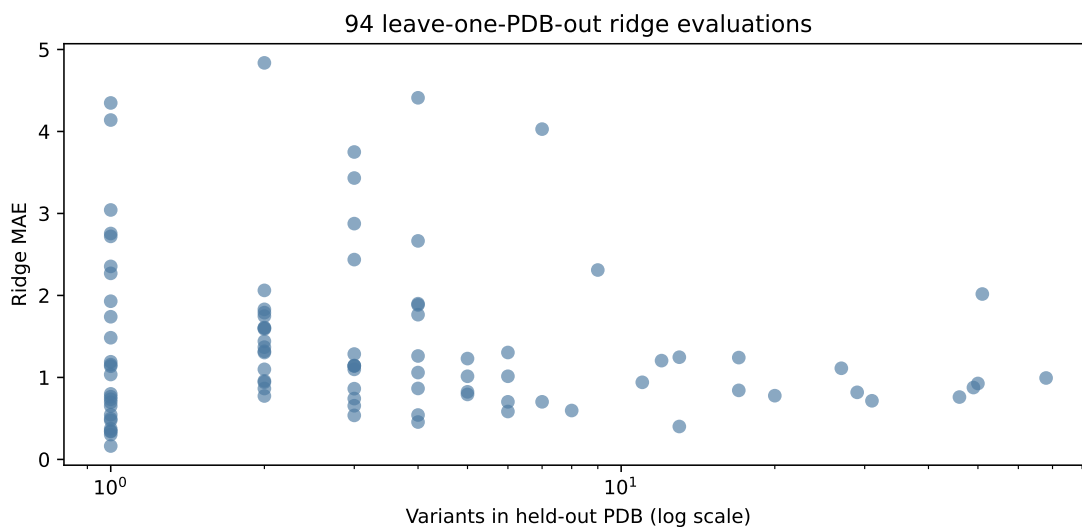


Figure 7: Leave-one-PDB-out ridge stress test. Each dot is one independently held-out PDB structure; the dashed and dotted lines use different weighting on exactly these predictions. This is not a GNN cross-validation figure.

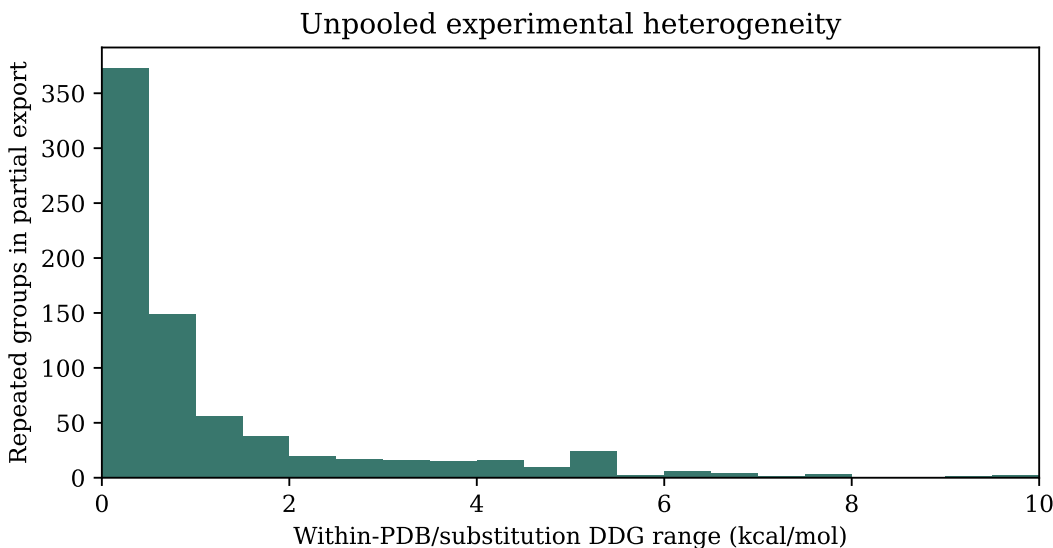


Figure 8: Observed DDG range within each repeated PDB/substitution group of a partial FireProtDB export. The tail is clipped visually at 10 kcal/mol; the numerical quantiles in the text use all records. No database-wide prevalence claim is made.

12 Disease target case boundaries

The RCSB entry 1TSR is a TP53 core-domain DNA complex; 2C9V is human SOD1; 1D5R is PTEN. Their presence does not establish that any specific clinically observed variant or rescue mutation is represented in our labeled benchmark. A source-grounded numbering hazard is concrete here: UniProt P00441 includes an initiator methionine absent from the mature SOD1 chain in PDB 2C9V. The literature A4V site is PDB mature residue 4, but UniProt canonical residue 5; G93A similarly maps to mature residue 93, canonical residue 94. TP53 1TSR maps canonical Y220 to chain A residue 220, and PTEN 1D5R maps canonical R130 to chain A residue 130. These were checked against retrieved UniProt FASTA sequences and observed structure residue identities, not inferred from variant names. The programmatic 19-site map is committed in the disease-site numbering JSON; all 19 selected sites match reference amino acids in the observed PDB chain. In 2C9V, mature SOD1 D90 and G93 are separated by 5.28 angstrom C-alpha distance, whereas A4 to G93 is 24.56 angstrom. In PTEN 1D5R, D107 to R130 measures 17.00 angstrom and R130 to R335 measures 28.60 angstrom. These are static wild-type coordinates, not mutant trajectory or rescue mechanisms. Literature on p53 Y220X cavities and SOD1 A4V/G93A structural biophysics motivates hypotheses but neither our graph score nor a generic positive stability delta demonstrates restored transcription, tumor suppression or motor-neuron survival. A future rescue nomination should state the disease variant, explicit second-site mutation, predicted stability margin, conformational mechanism, and independent functional test. No such nomination clears those checks yet.

13 Cross-database identity without mechanistic overreach

To avoid comparing different entities with similar names, we queried independent protein and variant databases. Ensembl REST resolved human TP53 to gene ENSG00000141510. cBioPortal independently resolved the cancer gene symbol to Entrez 7157, while NCBI dbSNP linked the ClinVar cross-reference to rs121912666. A UCSC hg38 RefSeq track query returned 15 overlapping transcript annotations around the ClinVar GRCh38 locus; overlapping transcripts are not 15 separate biological experiments. NCBI ClinVar E-utilities identified Y220C as VCV000127819, HGVS c.659A>G (p.Tyr220Cys), linked to a P04637 variant. A refined ChEMBL query

distinguished the human p53 target ChEMBL4096 linked to UniProt P04637 from mouse ChEMBL4164 and the many text matches for proteins merely named TP53-binding. PDBE returned the same 1TSR entry identity as RCSB, while InterPro and AlphaFold supplied TP53 protein and predicted-structure metadata. MaveDB carries a TP53 multiplexed functional score set but its listed naive Bayes score is a trained variant-effect integration, not direct thermodynamic DDG. These cross-checks prevent identity and evidence-type errors; they do not provide a measured Y220C double-mutant rescue. The exact record identifiers and raw API responses are committed in the repository results. MyVariant.info was queried but broad gene-text search returned many genomic records, so no variant-specific claim was drawn from that response.

14 Pathway context is not mutation rescue evidence

The scientific API audit extends beyond structural coordinates to test what would be needed for a functional claim. Reactome maps TP53 UniProt P04637 to 46 pathway records in the response we retrieved, including transcriptional and stress-response processes. QuickGO returned GO annotation records for P04637, but the 1,043-hit count includes mixed evidence and ontology terms rather than 1,043 independent assays of Y220C. KEGG hsa:7157, STRING and IntAct returned gene, network and interactor contexts. These annotations indicate relevant biology but cannot tell whether a second-site change restores a p53 transcriptional program. An OLS search for “TP53” under MONDO returned HGNC-symbol matches rather than a clean disease relation; we record the failed semantic match rather than presenting it as disease validation. ChEMBL target lookup likewise required species filtering: the first broad text query mixed p53-binding proteins with p53, and the refined human target is ChEMBL4096 linked to P04637. None of these services was used to manufacture a mutation-specific effect size. They supply identity checks and candidate endpoint selection for future experiments, while the rank-check tool continues to abstain from unsupported rescue decisions.

15 Observed mutant structures rather than assumed geometry

For TP53 Y220C, the wild-type DNA-complex PDB 1TSR cannot represent the physical mutant cavity by itself. We fetched actual experimental Y220C structures 8QWN, 6SHZ and 4AGO. After pairing identically numbered chain-A C-alpha atoms and fitting the positions of unchanged amino acids, their wild-type-to-mutant backbone RMSDs were 0.68, 0.54 and 0.57 angstrom; C-alpha displacements at position 220 were 0.32, 0.56 and 0.52 angstrom, respectively. For SOD1, experimentally observed A4V structures 1UXM and 3GZQ yielded backbone fits of 0.64 and 0.41 angstrom against 2C9V, while G93A structure 3GZO yielded 0.67 angstrom. These values, their fitting atom counts, and every per-residue C-alpha displacement are in `results/mutant_structure_panel.json`. The low global backbone displacements do not refute disease effects: missing side-chain volume, altered stability, metal binding, dynamics, DNA binding and experimental conditions are not captured by this single metric. The structures come from separate crystal contexts and are not an experimental double-mutant rescue test. This is a concrete step beyond mutating the amino-acid letter on a wild-type PDB without checking whether a mutant structure exists.

16 Published p53 rescue controls and model boundaries

A published p53 double-mutant study provides an unusually direct reality check for generic stability-based rescue claims (PMCID PMC305574). The authors measured G245S at $+1.22 \pm 0.22$ kcal/mol relative destabilization, N239Y at -1.37 ± 0.23 , and G245S/N239Y at 0.00 ± 0.22 under their experimental conditions. The double mutant therefore regained approximately wild-type folding stability. The simple additive prediction is $1.22 - 1.37 = -0.15$ kcal/mol; the observed-minus-additive interaction is $+0.15$ kcal/mol, small relative to the reported uncertainties. Yet stability is not interchangeable with DNA-binding activity. Their R249S variant showed $+1.69 \pm 0.21$ kcal/mol destabilization and no detectable binding in the reported assay, while H168R alone

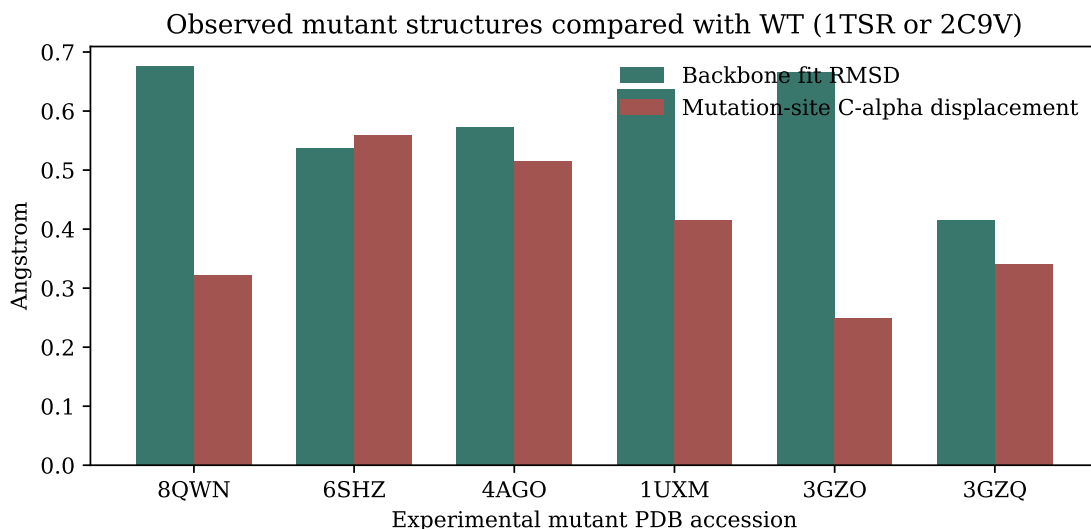


Figure 9: Observed mutant PDB chain-A C-alpha alignments to their wild-type references. Structural fits and site displacements are different measurements; neither measures thermodynamic stability or restored function.

showed $+2.75 \pm 0.21$ destabilization. R249S/H168R had $+1.72 \pm 0.21$ destabilization, almost no improvement versus R249S, but its measured half-binding concentration was 170 nM. The expected additive destabilization would be $1.69 + 2.75 = 4.44$ kcal/mol, a non-additive deviation of $1.72 - 4.44 = -2.72$ kcal/mol. The published values and source link are transcribed in the repository results. They are not experiments or discoveries from this work.

$$\epsilon_{AB} = \Delta\Delta G_{AB} - (\Delta\Delta G_A + \Delta\Delta G_B). \quad (16)$$

This interaction term makes a limitation of a single-mutation predictor explicit. Even a perfect marginal $\Delta\Delta G$ estimator would require a separate estimate of ϵ_{AB} to infer the double-mutant outcome. The R249S/H168R functional mechanism also depends on local loop conformation and DNA recognition. A single scalar stability prediction cannot certify functional rescue. A second published genetic study reports a 235-239-240 p53 suppressor motif that rescued 16 of 30 tested common cancer mutants in its yeast assay (DOI 10.1073/pnas.0401162101). It is a literature control, not a success attributable to this project. The Y220C candidate angle remains unvalidated in our pipeline; a credible nomination would require an explicit double-mutant model and independent binding or cell assay.

17 Measured wild-type p53 spatial context

The fetched 1TSR wild-type TP53 chain A supports a structural measurement independent of model training: C-alpha distances among named disease and suppressor sites. The committed coordinate calculation in `results/p53_contact_context.json` gives 8.73 angstrom between residues H168 and R249, consistent with their proximity in the published functional rescue mechanism. N239 to G245 is 7.98 angstrom and N239 to R273 is 7.64 angstrom; Y220 is substantially farther from the named motif, at 17.73 angstrom from N235 and more than 25 angstrom from R273. These numbers refer to C-alpha positions in the observed wild-type crystal structure, not altered side-chain contacts after mutation. They neither predict a double-mutant stability change nor establish a mechanistic connection to Y220C rescue. The geometry can nevertheless filter proposed local versus distal hypotheses: if a future claim depends on a direct contact between Y220 and N239 in 1TSR, it must be checked against the observed coordinates or a defensible alternate conformation, rather than simply invoking

the known suppressor motif.

$$d_{ij} = \|\mathbf{r}_i^{\text{CA}} - \mathbf{r}_j^{\text{CA}}\|_2. \quad (17)$$

18 Reproducibility and limitations

Install the package with pip, fetch accession structures separately, and run the benchmark for 15 epochs. Exact commands appear in the repository README. The data preparation and benchmark scripts carry no live calls during model training; network retrieval is an explicit separate step. Seventeen tests were passing at manuscript draft time; the testing status must be rechecked at publication. The package also offers a published-measurement double-mutant cycle calculator and an abstention check, in addition to residue graph inspection, benchmark training, group sensitivity, FireProtDB mapping and canonical-to-PDB site mapping. All raw S669 rows and the reduced FireProtDB sample retain source identifiers. A single train/test split cannot estimate run-to-run uncertainty; exact sequence deduplication alone does not enforce family-disjoint evaluation; a multi-seed, leave-protein-family-out evaluation remains necessary. The GNN architecture is intentionally modest, no published source has been rerun under matched training data, and no novel rescue mutation is experimentally verified. The benchmark is a negative result. No world record claim is made.

19 Interpretation checklist for follow-up experiments

A mutation-screening tool is useful only if it carries enough context to stop an invalid next step. Before any experiment, a prospective user should identify the parent disease variant in both canonical and construct/PDB numbering, the exact second-site residue and chain, whether the mutation is chemically accessible, and whether the model has actually seen a corresponding assay domain. For this study, TP53 Y220C, SOD1 A4V, and PTEN R130 variants were not measured by our GNN in a double-mutant setting. The correct answer to “has this tool shown rescue?” is no. To test TP53 Y220C compensation, a practical protocol would first measure folding stability and native-like occupancy of Y220C with and without one preregistered second-site change, then test sequence-specific DNA binding, cellular transcription and toxicity. All outcomes, including failure, need to be compared at the same temperature, pH, expression conditions and protein construct. That protocol is a proposed validation criterion; we have not performed it. For SOD1, an experiment must separately examine folding, dimerization and aggregation; a stability delta for one monomer does not establish ALS rescue. For PTEN, lipid phosphatase activity and membrane localization are separate from folding stability. These assay choices follow the biological role of each protein rather than treating numerical DDG as a clinical endpoint.

20 The abstention decision as a usable tool

Our negative GNN benchmark and literature p53 controls do not establish double-mutant rescue for these disease targets, so the tool includes an explicit rank-check command. It accepts a JSON record of evidence and refuses a mutation ranking when PDB accession, sequence alignment, residue numbering, DDG sign convention, representative training coverage or double-mutant assay support is missing. The bundled TP53 Y220C example has a verified structure and site numbering but no verified assay DDG convention, no relevant double-mutant training proteins and no target-specific validation. The command therefore returns `eligible:false` with four stated reasons. The default thresholds of 10 relevant training proteins and 20 mutation rows are conservative workflow settings, not validated statistical calibration or a predictive probability. Passing the check would be necessary but not sufficient for a biological rescue claim. This design makes negative evidence usable: it prevents the interface from turning a single-mutation score into an unsupported unwarranted recommendation. The guard is covered by tests for both incomplete and fully supplied input; a caller cannot override it by merely providing a promising

predicted number.

$$\text{eligible} = \bigwedge_{c \in \mathcal{C}} \mathbf{1}[c \text{ verified}], \quad \mathcal{C} = \{\text{identity, numbering, sign, coverage, double-mutant assay}\}. \quad (18)$$

21 Reproducibility checklist and failure ledger

The source manifest records URL and checksum for S669-linked PDBs. The corrected accession ledger records 163 used PDB structures: 157 unique S669 or partial FireProtDB mutation-data PDBs plus six disease-mutant structures directly compared with wild type. The FireProtDB bulk export did not finish; its partial state is a source limitation, not an example of a complete database census. A separate ThermoMutDB API query for a bounded 1-to-5 DDG interval returned 1,091 records, including repeated and multi-substitution records, so it was not pooled with the single-site S669 labels. The API query confirms source availability but does not repair the incomplete FireProtDB export. An external-service ledger in the committed results counts only services actually queried: twenty-five scientific data or literature surfaces, excluding GitHub and Drive infrastructure. It does not claim completion of a 40-tool external-use target. The first S669 benchmark attempt also revealed a JSON serialization failure when the constant train-mean predictor yielded undefined rank correlation. The stored code now emits null instead of a made-up rank metric. Another implementation flaw averaged over padded graph vertices; a test that pads the same graph to different sizes detected and prevented this bias. None of these fixes turns the negative published comparison into a win. The baseline configuration, seed, features, graph radius and edge cutoff are recorded in code and the machine-readable result. Raw structure downloads are not committed to Git because the accession files can be regenerated from the exact RCSB endpoints; the PDB SHA-256 ledger allows future users to notice if a file changes. Independent third-party predictor outputs are read from the S669 table and are not claimed to have been rerun.

22 Supplementary audit tables

The tables below expose measured groups and their uneven row counts. They are not new cohorts, independent replicates, or prospective rescue screens. Their machine-readable records and predictions are in the result JSON files. Numeric precision in tables is rounded for display; all summary calculations use stored full-precision values.

22.1 Leave-one-PDB-out ridge baseline

Each row fits a train-only ridge model excluding the named structure. The GNN was not retrained in this table.

PDB	Variants	Ridge MAE	Train-mean MAE
1A0F	1	1.191	0.840
1A7V	13	1.247	1.140
1BA3	4	0.542	0.853
1BFM	2	0.942	0.413
1BNL	1	0.347	0.141
1D5G	2	0.865	1.401
1DIV	50	0.926	1.253
1DXX	1	2.720	2.742
1EKG	4	1.765	1.170
1F8I	1	1.484	0.507
1FH5	1	1.137	1.564
1FRD	1	2.755	4.969
1FT8	13	0.401	0.648
1FXA	11	0.940	0.737
1G3P	27	1.112	0.790
1GLU	1	0.303	0.259
1GUA	49	0.877	0.980
1GWY	1	0.340	0.639
1H0X	2	1.830	1.301
1HCQ	2	1.319	1.082
1IOJ	1	2.354	1.364
1IR3	1	1.740	1.564
1ITM	3	3.750	2.915
1IV7	3	0.863	1.313
1IV9	1	4.139	5.269
1J8I	6	0.584	1.223
1JL9	1	1.155	0.479
1JLV	3	1.143	0.965
1L6H	3	0.537	0.556
1LVM	5	0.793	1.459
1N18	4	1.262	1.350
1N88	17	0.842	1.540
1NM1	2	1.304	0.510
1O1U	6	0.703	0.830
1O6X	20	0.777	0.581
1OSI	4	0.866	0.934
1PFL	5	1.014	1.590
1PRE	1	3.043	1.440
1PRG	7	0.702	0.903
1R2Y	1	4.348	3.243
1R6R	4	1.900	2.320
1SPD	2	1.591	1.416
1X0J	7	4.029	3.200
1XWS	4	2.665	2.556
1XXN	5	1.230	0.656

PDB	Variants	Ridge MAE	Train-mean MAE
1XZO	4	4.411	4.261
2ARF	1	0.698	0.245
2BJD	3	2.437	2.755
2C9Q	2	0.958	0.611
2CLR	4	1.885	1.817
2DVV	2	1.369	1.401
2H3F	1	2.269	2.035
2HBB	46	0.761	0.818
2JIE	51	2.017	1.921
2JUC	3	1.139	0.874
2KJ3	9	2.310	1.921
2KS4	2	1.608	1.481
2LTB	6	1.304	1.439
2M5S	2	1.602	2.044
2MPC	3	1.144	1.790
2N7Z	1	0.545	0.362
2NTE	2	1.790	1.192
2OUO	1	1.930	1.404
2PR5	17	1.242	1.669
2PTL	68	0.993	1.104
2RPN	2	2.062	1.687
2VY0	2	4.837	5.340
2WQG	29	0.818	0.800
2ZTA	8	0.596	0.630
3BCI	2	1.441	1.253
3BN0	1	0.651	0.545
3C2I	12	1.205	1.488
3D2A	3	3.432	3.802
3D3B	5	0.824	0.428
3DV0	31	0.715	0.920
3ECU	1	0.476	1.764
3FIS	1	0.374	0.462
3G1G	4	0.456	1.156
3K82	3	2.876	3.184
3L15	1	0.497	2.272
3MON	2	1.748	2.030
3O39	6	1.014	1.026
3S4M	4	1.059	0.801
3S92	1	0.734	2.996
4BJX	1	0.765	1.955
4BUQ	2	1.099	1.163
4HE7	3	0.655	0.752
4N6V	3	1.098	1.549
4WAA	3	0.742	0.897
4YEE	1	1.036	0.252
4YEF	1	0.799	0.032
5JXB	2	0.773	0.480
5OAQ	1	0.163	2.031
5VP3	3	1.285	1.096

22.2 Fixed test-set protein-group error

The signed difference is GNN MAE minus stored GeoDDG-Seq MAE on that PDB; positive means higher graph-model error. A one-variant row is not a reliable protein-wide comparison.

PDB	Variants	GNN MAE	GeoDDG-Seq MAE	Difference
1H0X	2	1.237	1.567	-0.330
1IOJ	1	2.367	0.229	+2.138
1ITM	3	2.802	2.890	-0.088
1JLV	3	1.080	1.486	-0.406
1N18	4	0.944	0.537	+0.406
1OSI	4	0.499	0.834	-0.335
1PRG	7	0.798	0.394	+0.405
1R6R	4	2.150	2.342	-0.192
1XXN	5	0.937	0.594	+0.343
1XZO	4	4.623	3.848	+0.775
2BJD	3	2.562	0.891	+1.672
2N7Z	1	0.301	1.423	-1.123
2NTE	2	1.836	1.337	+0.499
3BCI	2	1.024	1.379	-0.354
3BN0	1	0.594	0.597	-0.004
3FIS	1	0.464	1.037	-0.573
3L15	1	0.277	1.835	-1.558
3MON	2	1.573	1.710	-0.136
3O39	6	0.854	0.748	+0.105
3S92	1	1.213	4.179	-2.966
4BJX	1	1.852	1.729	+0.122
4BUQ	2	1.878	0.706	+1.172
4N6V	3	0.687	1.142	-0.455
4WAA	3	0.678	0.786	-0.108

22.3 Accession-backed use ledger

A structure enters this list only if an S669 graph used it, the partial FireProtDB audit directly matched mutation numbering, or a separate observed disease-mutant comparison used it. Multiple rows do not represent independent source studies. Full SHA-256 digests appear in the machine-readable ledger; short prefixes below are audit aids, not a substitute for full checksums.

PDB	S669 graph	FireProtDB direct	SHA-256 prefix
1A0F	yes	no	3bc0a7cf1534
1A5E	no	yes	295c74704ce0
1A7V	yes	no	7771652d2f71
1AM7	no	yes	a9cd7c656a88
1ANK	no	yes	7af4100057e8
1AON	no	yes	5cddcf0ebe22
1AZP	no	yes	3c1c11f8a625
1BA3	yes	no	a304920ee2ad
1BFM	yes	no	a60e18bb2ccf
1BNI	no	yes	e7a0612d97d3
1BNL	yes	no	6ba596ec06d6
1C9O	no	yes	202f922b44d8
1CEY	no	yes	7fc320d94e2c
1CLW	no	yes	c56ea831458c
1CSP	no	yes	0b3335beb321
1D5G	yes	no	00caca169656
1DIL	no	yes	1ed773b76af1
1DIV	yes	yes	5bc0f0e1fbf7
1DKT	no	yes	aa5b83105ee2
1DPM	no	yes	ea3214353cf2
1DXX	yes	yes	8daac296a887
1EKG	yes	yes	f9a19c74c785
1EL1	no	yes	dfec955486c5
1F8I	yes	no	ee75cf1b376b
1FH5	yes	no	d91231bc3b14
1FRD	yes	no	0d0894d9e57a
1FT8	yes	no	987bf04553a9
1FXA	yes	no	fb8d8c49b72e
1G3P	yes	no	e17cea0deace
1GLU	yes	no	17f9b479a641
1GUA	yes	no	f7122b06f9d8
1GWY	yes	yes	73679a8d14d8
1H0X	yes	no	df5650ec5125
1HCQ	yes	no	6a9f68a71836
1HFY	no	yes	25bc80d6ebd5
1HK0	no	yes	dda174b0c4b6
1HUE	no	yes	b89a0d9e4ffd
1IGV	no	yes	9a5228eecf3d
1IMQ	no	yes	139b46f0c693
1IO2	no	yes	8fc92c7be66f
1IOJ	yes	no	cac7f60005a6
1IR3	yes	no	83f1a55aa06d
1IRO	no	yes	95069c12563c
1ITM	yes	no	94179df65dc0
1IV7	yes	no	85c8fe587b3f
1IV9	yes	no	3bf754137550
1J8I	yes	no	411da189ad0a
1JL9	yes	no	f6660cbb6939
1JLV	yes	no	cde6ff76ccce
1JNX	no	yes	8d932403fc31

PDB	S669 graph	FireProtDB direct	SHA-256 prefix
1K85	no	yes	991c1012f5e0
1L6H	yes	no	f87c88d070d5
1LVM	yes	no	a11497fd324f
1LZ1	no	yes	19ca7f7e9788
1MBG	no	yes	a23a05fbfe93
1MJC	no	yes	f945636068ad
1MSI	no	yes	05044e49e3c7
1N18	yes	no	39aceb1a1ac0
1N88	yes	no	c7c30c565a7f
1NM1	yes	no	3310a8cff33a
1O1U	yes	no	e81a301b6823
1O6X	yes	no	ff5ae0ce0a36
1OH0	no	yes	79a37b665c53
1OIA	no	yes	0ac129d79c03
1ONC	no	yes	c25c67b82726
1OSI	yes	no	a59df955fc0d
1PFL	yes	no	9dbf69adb002
1PRE	yes	no	6ffc49adc15f
1PRG	yes	no	66fa620191fc
1QGV	no	yes	61c5823f9411
1QLP	no	yes	d6368c73af80
1R2Y	yes	no	50c3b3e7c985
1R6R	yes	yes	41434d9cd4ec
1RGG	no	yes	221692d5d6c4
1RIS	no	yes	e7c5b65055b4
1ROP	no	yes	974fcb647055
1SCE	no	yes	1efe4b7cf4aa
1SPD	yes	no	bc1047c32a42
1STN	no	yes	ef725bf74ca7
1TEN	no	yes	5b2165e4e495
1UXM	no	no	6f2ac4056387
1W4H	no	yes	7f57ff58ef4b
1WQ5	no	yes	a6ea3025f34f
1X0J	yes	no	806b49035ab0
1XWS	yes	yes	c6ba50b25016
1XXN	yes	no	fee7b53ef3b1
1XZO	yes	no	2d2bd578e191
1Y4Y	no	yes	56c194a7dafb
2A36	no	yes	2c0ea5f5c121
2ADA	no	yes	d0ffcd9a18c
2ARF	yes	yes	411774993504
2BJD	yes	no	c96f0fbb8a51
2BRD	no	yes	d4e491572ad9
2C9Q	yes	no	255e1e7e7547
2CI2	no	yes	8a42f06a0a59
2CLR	yes	no	ee2d2315b5be
2CRK	no	yes	db86ddbbae8e4
2DVV	yes	no	c186f5dbbebe
2H3F	yes	no	1d799bdb3919
2HBB	yes	no	857c21059c7f
2JIE	yes	no	cab04901fd62
2JUC	yes	no	b7a82e571dd4
2JYS	no	yes	e4083c342db2
2K7J	no	yes	97fc095d1ea1
2KJ3	yes	yes	4f3d860b3f60
2KS4	yes	yes	227656214e1e
2LCP	no	yes	226e7fb14161
2LSB	no	yes	cbc966516042

PDB	S669 graph	FireProtDB direct	SHA-256 prefix
2LTB	yes	no	cbfe5044bb22
2M5S	yes	no	fd00fcac0d2b
2MPC	yes	no	98b933c8c317
2N7Z	yes	no	b432f9863ef6
2NTE	yes	no	afd8fdb90873
2OOU	yes	no	bfccda360a4f
2PR5	yes	no	0734a61064e4
2PTL	yes	no	f56503afc7c5
2RN2	no	yes	fedbb7b49741
2RPN	yes	no	da39dafa88ff
2TS1	no	yes	e9eb279f61e4
2V0U	no	yes	f112a2f2505f
2VY0	yes	no	c0de6aec03dc
2W0K	no	yes	49be71ae13c9
2WQG	yes	no	7fb8f6de00ac
2WSY	no	yes	1cedf7f76be5
2ZTA	yes	no	c71d709f5d11
3BCI	yes	no	a4a83ea40029
3BDN	no	yes	184d751449a3
3BN0	yes	no	65532372afa3
3C2I	yes	no	f96ee9192eab
3D2A	yes	no	a051ddefca14
3D3B	yes	no	1ed3bc2176e7
3DV0	yes	no	d7f621391f05
3ECU	yes	no	c8cc3ea0abe4
3FIS	yes	no	cb64106e5575
3G1G	yes	no	a701c2462ab8
3GZO	no	no	e91f50fb04cd
3GZQ	no	no	657c0cd1da57
3K82	yes	yes	7a47105ff58c
3L15	yes	no	1c77fe2aef98
3MON	yes	no	3ec8a943bdcd
3MXF	no	yes	534dcb8ea9fa
3O39	yes	no	58c1a42b5c75
3PGK	no	yes	32abf08811f5
3S4M	yes	no	0e5ae2d53c91
3S92	yes	yes	b8eb3dbf535a
4AGO	no	no	07388d55c56b
4BJX	yes	no	7835025ffa5f
4BUQ	yes	no	d200fc9cfc6d
4HE7	yes	no	0f87c1998c9e
4LYZ	no	yes	869119f51d45
4N6V	yes	no	5a8020485843
4WAA	yes	no	c49fe7159fc2
4YEE	yes	no	ee54d35d50dc
4YEF	yes	no	2d5783795dd8
5AZU	no	yes	ce1a9ce09e68
5CG0	no	yes	ef1c85d1298b
5DFR	no	yes	52dc93a35ce2
5JXB	yes	no	605b00b48e6a
5OAQ	yes	no	158f2572d527
5VP3	yes	no	a7022fc3b652
6BQG	no	yes	92802a710045
6SHZ	no	no	c21f92ede305
8QWN	no	no	7617ae6a2f9a

22.4 What these counts cannot establish

The accession total counts structures used by distinct computational analyses, not distinct biological experiments. A PDB can contain variants with shared sequence background, assay labels can derive from related publications, and S669 predictor columns may have unknown historical training overlap with these proteins. FireProtDB is a partial download with a direct-numbering screen, not an exhaustively harmonized stability dataset. The fixed protein holdout cannot by itself justify a disease-rescue nomination. Published p53 suppressor evidence shows that functional rescue may happen with little stability gain, so a future follow-up needs disease-mutant-specific double-mutant measurements and functional endpoints. The guarded rank-check command deliberately abstains in their absence.

23 Source ledger

Full machine-readable source locations are preserved in data/SOURCES.md. The S669 benchmark table was obtained from the GeoStab GitHub repository at <https://github.com/Gonglab-THU/GeoStab>; the original release is Zenodo record <https://zenodo.org/records/7568094>. The original benchmark DOI is 10.1093/bib/bbab555. Protein structures are archived at <https://www.rcsb.org/>, and UniProt target sequences at <https://www.uniprot.org/>. Relevant entries are RCSB 1TSR, 2C9V, 1D5R and UniProt P04637, P00441, P60484. FireProtDB was obtained from <https://loschmidt.chemi.muni.cz/fireprotdb/download/>. Its DDG sign definition is documented at <https://loschmidt.chemi.muni.cz/fireprotdb/help/>. The p53 structural review is PMC7307883; SOD1 A4V/G93A biophysical work is PMC2787720 (public URLs in the repository source ledger). These sources motivate interpretation but are not experimental validation of a mutation proposed by this study.